

# Maps

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Data Structures  
MeerMans

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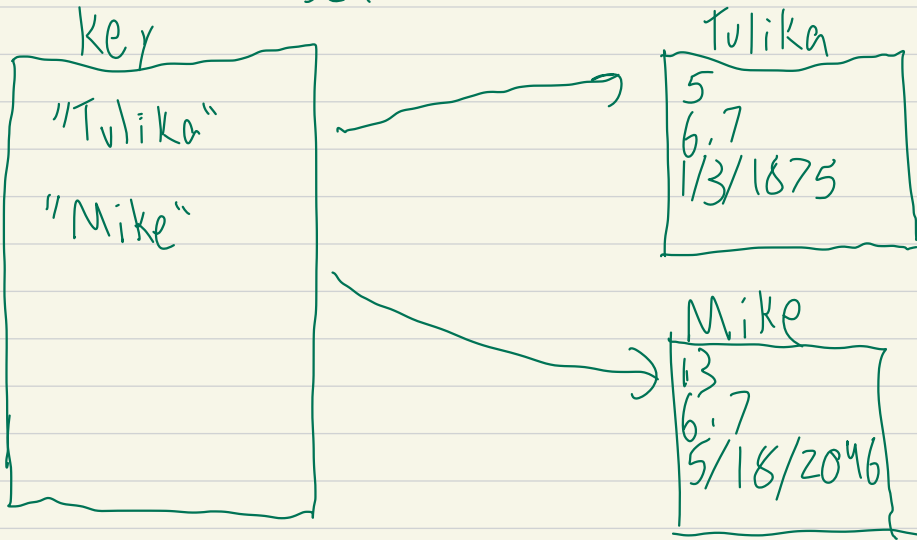
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Map a key to a value



Set of  
values in  
a set



# Map Methods

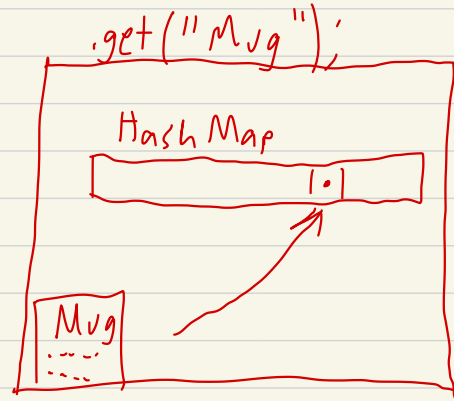
Keys are unique

$\text{Map} < K, V >$

Looking up a Key gives you a Value

HashMap: Keys are in a HashSet

TreeMap: Keys are in a TreeSet



Syntax:

`Map<__, __> VarName = new __ Map<>();`

`Map<String, DictionaryEntry> mW = new HashMap<>();`

Methods:

V put(key, value) → add a new pair, return old value

int size() → # of keys

boolean containsKey(key) → t/f if key is in Map

boolean containsValue(value)\* → t/f if value is in Map

V get(key) → returns a value at the key or null

V getOrDefault(key, value) → same as get() but specify what to default to if the key is not in the map

V remove(key) → returns the key/value pair

Set<K> keySet() → returns a set of all of your keys

Collection Values() → returns all of the values

Set<Entry<K, V>> entrySet → Returns a set of all of your entries (both the key and value)